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Solving Vehicle Routing Problem: A Big Data Analytic Approach

SHAOQING ZHENG
Hangzhou College of Commerce, Zhejiang Gongshang University, Hangzhou 311599, China
e-mail: sq_zheng@yeah.net

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ABSTRACT In the transport industry, the cost effectiveness relies heavily on the rational design of the transport routes. However, the traditional theories and methods on vehicle routing problem (VRP) cannot describe the dynamic features of travel time accurately. To solve the problem, this paper puts forward a dynamic VRP model based on big data analysis on traffic flow, and solves it by the genetic algorithm (GA). It is assumed that the real-time traffic data are updated every 15mins in the transport network, and the customer demand is updated dynamically from time to time. The example analysis shows that my model and its route adjustment strategy can minimize the total transport cost by routing the vehicles from multiple depots under the soft time window. The research findings help transport enterprises to make effective use of vehicles and receive more profits.

INDEX TERMS Big data analysis, vehicle routing problem (VRP), genetic algorithm (GA), minimum transport cost, vehicle sharing.

I. INTRODUCTION

In the transport industry, the cost effectiveness relies heavily on the rational design of the transport routes. A rational routing plan enables vehicles to complete the transport tasks at the minimum cost and time, covering the shortest distance [1]–[6]. For each vehicle, the travel distance is the route length between the start and end points. To develop a suitable routing plan, the set of routes must be optimized for a fleet of vehicles to traverse in order to deliver to a given set of customers [7].

The vehicle routing problem (VRP) was proposed by Dantzig and Ramser in 1959 [8], and has been solved by various mathematical models. In the VRP, the travel time between any start and end points as a constant. For each vehicle, the travel time varies with road sections, due to the nonuniform state in actual operations. For example, the travel time is affected by the weather and traffic condition. Despite the local variations in travel time, the route design and operation sequence remain unchanged. The traditional VRP theories and methods cannot describe the dynamic features of travel time accurately, calling for a dynamic VRP strategy.

Therefore, this paper puts forward a dynamic VRP model and solves it by genetic algorithm (GA). My model and its route adjustment strategy were verified through example analysis.

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II. LITERATURE REVIEW

There are four types of inputs to the VRP, namely, depot information, customer information, road information and vehicle information. The existing studies on dynamic VRP mainly deal with customer information and road information [9]–[11].

The dynamics of customer information are reflected by the time-variation in the number and demand of customers [12], [13]. Pavone *et al.* [14] proposed a service strategy for the VRP with variable number of customers, and proved its superiority over the first come, first serve strategy. Barkaoui *et al.* [15] explored the VRP with a fixed number of customers and variable customer demand.

The dynamics of road information are resulted from the changes of a vehicle's travel time on a road section. These changes take place due to the variation in traffic flow and weather, the occurrence of traffic accidents, etc. In the VRP, the travel time of a vehicle between two adjacent customers is determined on not only the distance between these customers, but also the time the vehicle leaving from the previous customer. The variation in travel time indirectly affects the customer sequence served by the vehicle. Balseiro *et al.* [16] designed a position-dependent VRP, whose cost (time) depends on the position of the departure customer in the route,

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rather than the time of departure. Kuo [17] introduced a time-dependent VRP, in which the travel cost (time) varies with road sections, described the problem with a mixed integer linear programming model, and adopted a heuristic algorithm to solve the model. Kazemian *et al.* [18] explored the VRP whose travel time in road sections does not obey the first in, first out rule, described the speed of vehicles in road sections as a step function, and derived the travel time in road sections from vehicle speed and travel distance. Kok *et al.* [19] determined the vehicle speed for each road section based on the location, grade and direction of the section. Nejad *et al.* [20] developed a route adjustment strategy under the uncertain travel time of a road section. Targeting the VRP with frequent and accident congestions, Ye *et al.* [21] arranged the initial routes based on the travel time under frequent congestion, updated the instantaneous traffic information at intersections, and proved that this update strategy is better than timing updating and customer updating.

In the light of the above, this paper considers that the real-time traffic data are updated every 15mins in the transport network, and the customer demand is updated dynamically from time to time. Then, the objective of the VRP is to minimize the total cost by routing the vehicles from multiple depots under the time window constraints.

III. DYNAMIC VRP MODEL

In the vehicle routing problem, the dynamic vehicle routing problem is the most basic component, but the value of these elements is no longer a fixed value, but changes with time. In dynamic vehicle routing problem, the attribute relationship among vehicle, customer and road network will change with time.

The VRP model needs to serve a set of customers, either concentrated in the same place or scattered in different places, with a fleet of vehicles. The items for each customer must be delivered in one visit by one vehicle, and each vehicle should return to the start point after completing its delivery task.

A. STANDARD VRP AND MATHEMATICAL MODEL

In the standard VRP, all customers should be visited through a rational route network, and the transport cost should be minimized under constraints on travel time, vehicle load and time window. The transport cost is related to various factors, including the total number and mileage of vehicles.

The standard VRP is essentially a route optimization problem with loading capacity. Drawing on graph theory, the problem can illustrated as a complete graph [22] $G = (V, E)$, where $V = \{0, 1, 2, \ldots, n\}$ is a set of nodes (depots and customers) and $E = \{(i, j)|i, j \in V, i \neq j\}$ is a set of edges (routes). The set of customers and the set of vehicles can be expressed as $V' = \{0, 1, 2, \ldots n\}$ and $K = \{0, 1, 2, \ldots, m\}$, respectively. In addition, the number of required vehicles is denoted as M, the maximum load of a vehicle as Q, the demand of customer i as d_i, the transport cost via route (i, j) as v_{ij}, and the fact that vehicle k travels across route (i, j) as x_{ijk}.

The standard VRP model has a hierarchical objective function. In general, the first objective is to minimize the number of vehicles, and the second objective is to minimize the transport cost of all vehicles. The objective function $F(x)$ can be expressed as:

$$min\ F(x) = s_1 M + s_2 \sum_{k=1}^{M} \sum_{i=0}^{N} \sum_{j=0}^{N} v_{ij} x_{ijk} \quad (1)$$

where, s_1 and s_2 are two integers ($s_1 \gg s_2$) that ensures the pursuit of the two objectives.

The standard VRP model is subjected to the following constraints:

$$\sum_{i=0}^{N} \sum_{k=1}^{M} x_{ijk} = 1, \quad \forall j = 1, 2, \ldots, N \quad (2)$$

$$\sum_{j=0}^{N} \sum_{k=1}^{M} x_{ijk} = 1, \quad \forall i = 1, 2, \ldots, N \quad (3)$$

$$\sum_{k=1}^{M} \sum_{i=1}^{N} x_{i0k} = \sum_{k=1}^{M} \sum_{j=1}^{N} x_{0jk} = M \quad (4)$$

$$\sum_{i=0}^{N} b_i \left(\sum_{j=0}^{N} x_{ijk} \right) \leq Q, \quad \forall k = 1, 2, \ldots, M \quad (5)$$

$$\sum_{i \in S} \sum_{j \in S} x_{ijk} \geq 1 \quad (6)$$

where, N represents the number of vehicles, M represents the number of customers.

Constraints (2) and (3) ensure that each customer is served only once by one vehicle. Constraint (4) ensures that each vehicle leaves from a depot and returns to that depot after completing its delivery task. Constraint (5) ensures that the items loaded onto a vehicle do not exceed the loading capacity of that vehicle. Constraint (6) eliminates the presence of secondary path.

The standard VRP model shows that roads, customers, depots, vehicles and objective(s) are the core elements of the problem. Specifically, roads lay the basis for item delivery; customers are nodes in the route network and should be visited within time limits; depots are the start and end points of vehicles; vehicles are responsible for picking up and delivering items to costumers; the objective(s) may be singular or plural depending on the actual situation.

If there is only one depot, the VRP is a single-depot problem; if there are multiple depots, the VRP is a multi-depot problem. Similarly, the VRP may or may not have time window, according to the actual situation. If the vehicle information is determined in advance, then the routing plan is already optimal, making the problem a static VRP. Otherwise, the routing plan needs to be optimized continuous in the operations, making the problem a dynamic VRP.

B. DYNAMIC VRP AND MATHEMATICAL MODEL

The dynamic VRP with vehicle sharing and real-time traffic information can be described as follows: There are N customers to be served by have K vehicles from M depots. For each customer, the demand is b_i and the service time window is $[ET_i, LT_i]$. For each vehicle, the maximum load is Q, the loading/unloading time at the depot is t_1, and the loading/unloading time at each customer is t_2. The aim of the

problem is to prepare a routing plan according to real-time traffic information that minimizes the total transport cost under the time windows and other constraints.

Constraint 1: Each vehicle should leave from a depot and return to the depot after completing the delivery task.

Constraint 2: The items for each customer must be delivered in one visit by one vehicle.

Constraint 3: In a planned period, the travel time of each vehicle should not exceed the maximum travel time, and the travel distance of each vehicle should not surpass the maximum travel distance.

Before setting up the mathematical model of the dynamic VRP, the following parameters must be defined: the total number of customers, N; the total number of vehicles, K; the loading capacity of each vehicle, Q; the maximum travel time of each vehicle, T; the maximum travel distance of each vehicle, D; the fixed cost of each vehicle, A_r; the transport cost per unit time of each vehicle, A_f; the actual travel time for vehicle k to move from customer i to customer j, t_{ijmk}; the travel time for vehicle k to move from customer i to customer j under the condition of free flow, t_{0ijmk}; the traffic capacity of the road section from customer i to customer j, c_{ij}; the actual traffic volume of of the road section from customer i to customer j, x_{ij}; the penalty on a vehicle for failing to meet the time window, $R(t)$; the moment that vehicle k of depot m reaches customer n, t_{nmk}; the demand of customer n to be served by vehicle k of depot m, q_{nmk}; the loading/unloading time of vehicle k of depot m at the depot, wt_{0mk}; the loading/unloading time of vehicle k of depot m at customer n, wt_{nmk}.

On this basis, the decision variables can be defined as:

$$x_{ijmk} = \begin{cases} 1 & \textit{vehicle k of depot m travels from customer} \\ & \textit{i to customer j} \\ 0 & \textit{otherwise} \end{cases}$$

$$y_{nmk} = \begin{cases} 1 & \textit{vehicle k of depot n serves customer n} \\ 0 & \textit{otherwise} \end{cases}$$

Then, the initial routing plan and model can be established based on the known information about customers and static traffic. Since the service time window is a soft time window, a penalty function $P()$ was designed to punish the vehicle failing to meet the time window. The penalty function $P()$ has the following relationship with the service time t:

$$P(t) = \begin{cases} R, & t < S \\ \alpha \times (ET_i - t), & S \le t \le ET_i \\ 0, & ET_i \le t \le L \\ \beta \times (t - LT_i), & LT_i \le t \le L \\ R, & t > L \end{cases} \tag{7}$$

where α and β are adjusting parameters; R is an infinite penalty; S and L are the start and last moments to accept the service, respectively; ET_i and LT_i are the earliest and latest moments of the time window, respectively.

If the vehicle reached the customer earlier than S, the customer will not accept the items and the vehicle will face an infinite penalty R. If the vehicle reached the customer between S and ET_i, the customer will accept the items and the vehicle will face a penalty, whose value depends on the time of arrival. If the vehicle reached the customer between ET_i and LT_i, the customer will accept the items and the vehicle will not face any penalty. If the vehicle reached the customer between LT_i and L, the customer will accept the items and the vehicle will face a penalty, whose value depends on the time of arrival. If the vehicle reached the customer after L, the customer will not accept the items and the vehicle will face an infinite penalty R.

As mentioned before, the real-time traffic data are updated every 15mins in the transport network, and the customer demand is updated dynamically from time to time. Therefore, the routing plan must be updated constantly to fit the latest situation. Before the update, the vehicles have already served some customers, and unloaded some of the items. Hence, new customers can be added to the routes based on the real-time traffic situation and the spare capacity of vehicles. The vehicles should be rearranged if the existing ones cannot satisfy the demand of new customers.

Let Q_e be the spare load capacity of each vehicle, P be the number of vehicles in operation, and U be the total number of unserved customers and new customers. Then, the real-time route optimization model can be established as:

$$\begin{aligned} min \Big\{ & A_r \cdot \sum_{m=1}^{M} \sum_{n=1}^{U+M} \sum_{k=1}^{P_m} y_{nmk} \\ & + A_f \cdot \sum_{m=1}^{M} \sum_{i,j=1}^{U+M} \sum_{k=1}^{P_m} x_{ijmk} \cdot t_{ij} \\ & + \sum_{m=1}^{M} \sum_{i=1}^{U} \sum_{k=1}^{K_m+P_m} \int P(t)\, d_t \Big\} \end{aligned} \tag{8}$$

$$\text{s.t.} \sum_{n=1}^{N} b_n \cdot y_{nmk} \le Q, \quad k \in \{1, 2, \ldots, k_m\}, \quad m \in 1, 2, \ldots, M \tag{9}$$

$$\sum_{m=1}^{M} \sum_{i=1}^{M+N} \sum_{j=1}^{M+N} x_{ijmt} \le D, \quad t \in \{1, 2, \ldots, t_m\}, \quad m \in \{1, 2, \ldots, M\} \tag{10}$$

$$\sum_{i=1}^{M+N} x_{ijmt} = y_{jmt}, \quad j \in \{1, 2, \ldots, N\} \tag{11}$$

$$\sum_{j=1}^{M+N} x_{ijmt} = y_{imt}, \quad i \in \{1, 2, \ldots, N\} \tag{12}$$

$$\sum_{m=1}^{M} \sum_{t=1}^{t_m} y_{nmt} = 1, \quad n \in \{1, 2, \ldots, N\} \tag{13}$$

$$\sum_{n=1}^{N} b_{nmt} = Q, t \in \{1, 2, \ldots, t_m\}, \quad m \in 1, 2, \ldots, M \tag{14}$$

$$\sum_{i=1}^{M+N} x_{inmt} \cdot (b_{nmt} - b_n) \ge 0, n \in \{1, 2, \ldots, N\}, \quad t \in \{1, 2, \ldots, t_m\}, \ m \in 1, 2, \ldots, M \tag{15}$$

Equation (8) is the objective function, which represents the minimum total transport cost. Constraint (9) ensures that the items loaded onto a vehicle does not exceed the loading capacity of that vehicle. Constraint (10) ensures that each vehicle does not surpass the maximum travel distance.

TABLE 1. Results of chromosome decoding.

Route No.	Departure Distribution Center	Original Distribution Center	Service Order	Returned Distribution Center
Route 1	1	1	12-17-25-18	3
Route 2	1	1	9-3-10-15-24	2
Route 3	2	2	8-13-14-1-19	2
Route 4	2	3	2-20-23-7	1
Route 5	3	3	6-11-4-5	2
Route 6	3	1	21-16-22	3

Constraint (11) ensures that each vehicle can only visit each customer only. Constraints (12) $\sim$ (14) ensure that each customer can only be served by one vehicle. Constraint (15) ensure that each vehicle leaves from the depot at full load.

C. ALGORITHM DESIGN FOR MODEL SOLUTION

Considering the features of the dynamic VRP, the GA was selected [23]–[29] to solve my model. The M depots were coded with natural numbers as 1,000, 2,000, 3,000...M,000. Then, the vehicles of depot M were coded as $k_1 + k_2 + \ldots + k_{m-1} + 1, \ldots, \sum_{m=1}^{M} k_m$.

Taking 3 depots, 2 vehicles of each depot and 20 customers for example, the chromosome decoding can be illustrated in Table 1.

During the selection operation, the individuals with large fitness are more likely to be retained. Therefore, the objective function (minimizing the transport cost) of the dynamic VRP should be transformed as:

$$f_i = \frac{1}{Z_i} \tag{16}$$

The selection is followed by the crossover operation, in which the genes of two individuals are partly swapped to form new individuals. Here, the sequential crossover method is adopted to retain and fuse the ordered structural units of different chromosomes. Taking the two parent chromosomes q_1 and q_2 of eight routes for example, the crossover points were selected randomly as follows:

$$q_1 : (1\ 2\ 3\ 4\ 5\ 6 \mid 7\ 8)$$
$$q_2 : (3\ 4\ 8\ 5\ 2\ 7 \mid 1\ 6)$$

During the crossover, the customers [1, 6] were identified and emptied in the parent chromosomes, producing two child chromosomes o_1 and o_2:

$$o_1 : (6\ 2\ 3\ 4\ 5\ 1 \mid 7\ 8)$$
$$o_2 : (3\ 4\ 8\ 5\ 7\ 2 \mid 16)$$

IV. SIMULATION AND RESULTS ANALYSIS

This section verifies the effectiveness of my model through simulation. To begin with, the route network was treated as 10∗10 square grids, forming 100 intersections and 180 sections. The real-time traffic flow was updated every 15mins. The travel time on each section was initialized through big data analysis. A total of 25 customers were simulated. Five of them have dynamic demands, and twenty have static demands. The information of these customers is given in Tables 2 and 3 below.

TABLE 2. Information of static customers.

Customer No.	Position	Demand	Time window
1	(3,2)	1.4	(70,100)
2	(9,3)	2.1	(70,90)
3	(2,7)	1.4	(80,110)
4	(9,2)	2.8	(30,60)
5	(5.6)	0.7	(10,50)
6	(8,10)	1.4	(60,100)
7	(5,5)	0.7	(70,100)
8	(5,4)	1.4	(30,70)
9	(9,7)	2.8	(70,120)
10	(5,9)	0.7	(50,80)
11	(1,9)	3.5	(80,120)
12	(7,8)	1.4	(20,60)
13	(3,9)	1.4	(10,50)
14	(7,3)	2.8	(40,90)
15	(1,6)	0.7	(30,100)
16	(5,8)	1.4	(40,70)
17	(7,4)	0.7	(50,90)
18	(8,2)	3.5	(40,90)
19	(8,1)	2.1	(50,110)
20	(5,3)	2.1	(60,120)

TABLE 3. Information of dynamic customers.

Customer No.	Position	Demand	Time window	Demand Moment
21	(7,5)	3.5	(40,150)	20
22	(3,8)	2.1	(120,180)	40
23	(9,1)	2.1	(200,300)	70
24	(6,2)	2.8	(300,420)	90
25	(7.6)	3.5	(120,250)	100

For each vehicle, it is assumed that the load capacity is 10ton, the maximum travel distance is 150km, the maximum travel time is 8h, the transport cost per unit time is 20yuan/h, and the fixed cost is 180 yuan. Each depot operates in the period of [0h, 10h].

The proposed GA was employed to generate the initial routing plan based on the known information on static customers and dynamic traffic. The GA parameters were configured as: population size, 200; crossover probability, 0.8; mutation probability, 0.2; maximum number of iterations, 300.

Through multiple MATLAB simulations, the routing plan with the minimal total transport cost (3,615.78 yuan) was selected as the initial optimal routing plan (Table 4), including the start and end depot, the total demand on the route, the travel time, travel distance, and penalty for failing to meet time window of each vehicle. The initial routes for the vehicles are presented as Figure 1.

As shown in Table 4, the initial routing plan uses six vehicles from three depots. For each vehicle, total demand on the route should be controlled within the loading capacity, the total travel time should not exceed the maximum travel time, the total travel distance should not surpass the maximum travel distance, and the service outside the time window should be penalized.

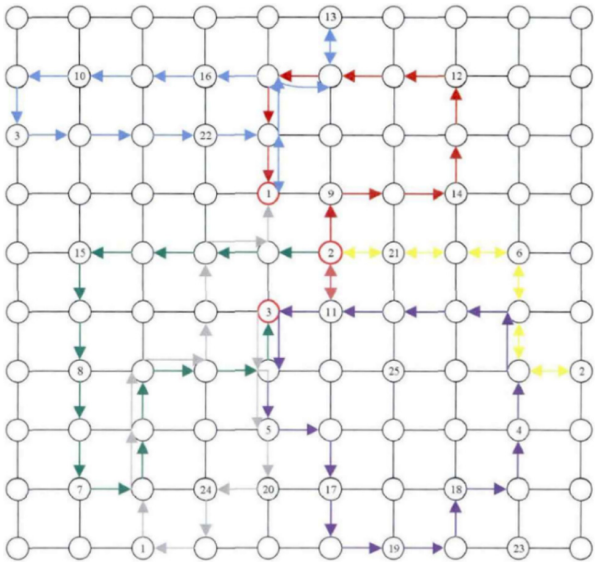


FIGURE 1. The initial routes.

TABLE 4. The initial routing plan.

Vehicle No.	Driving time	Demand	Driving distance	Time penalty
1	5:52	7.7	110	142.56
2	7:33	9.1	140	387.21
3	3:21	3.5	90	0
4	5:45	5.6	120	128.24
5	7:28	9.1	150	478.51
6	6:05	8.4	130	0

Since the real-time traffic flow is updated every 15mins, the travel time of a new road section can be derived through the big data analysis on traffic flow. When five dynamic customers emerged, route adjustment should be made at four time points: T = 20, T = 40, T = 70 and T = 90.

Taking T = 20 for instance, customer 21 was added at this time point. According to the randomly generated traffic flow of each road section, vehicle 3 was arranged to serve this customer along the route: depot 2-21-6-2-depot 2. The total transport cost of this route was 276.5 yuan. The adjusted routes to serve the five dynamic customers are shown in Figure 2.

Based on the adjustments at the four time points, the final optimized routing plan is given in Table 5.

If the initial distribution plan is still used for distribution, but the distribution route is not updated in real time. And for new customers, if there are idle vehicles, and the remaining goods meet the needs of customers, they will be distributed, otherwise a vehicle car will be dispatched from the distribution center for distribution. The comparative scheme is to dispatch 8 vehicles at a total cost of 4528.72. The results are compared as shown in Table 6.

For large scale data, we selected data sets of 23 vehicles and 120 distribution centers. The results are compared as shown in Table 7.

In order to verify the effectiveness of the proposed algorithm, we compared the proposed algorithm with the exact algorithm using CPLEX, which can represent complex business problems as mathematical programming models. The comparison results are shown in Table 8 and Table 9, respectively.

From Table 8 and Table 9, we can see that the proposed algorithm in this paper is better than algorithm using CPLEX.

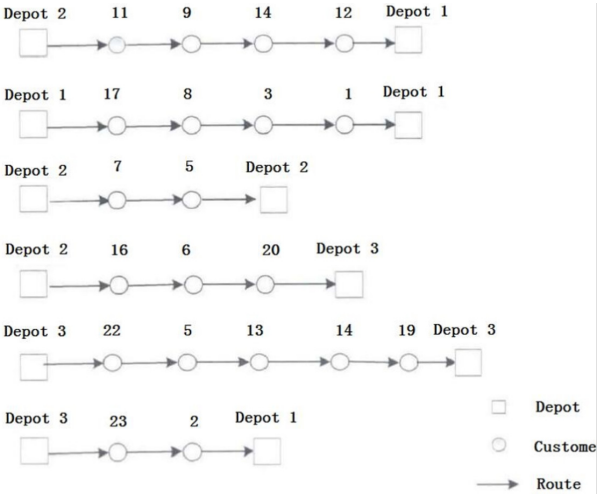


FIGURE 2. Adjusted routes for vehicles at T = 20.

TABLE 5. The final optimized routing plan.

Vehicle No.	Route	Demand	Cost	Time penalty
1	Depot 2-7-9-21-17-Depot 1	7.7	467.21	142.56
2	Depot 1-15-8-6-10-Depot 1	9.1	984.23	387.21
3	Depot 3-11-4-25-Depot 2	3.5	469.72	0
4	Depot 2-5-22-24-Depot 2	5.6	433.98	128.24
5	Depot 2-3-13-16-19-Depot 3	9.1	866.19	478.51
6	Depot 3-18-12-23-Depot 1	8.4	337.87	0
7	Depot 3-2-14-1-20-Depot 3	8.4	319.62	0

TABLE 6. Comparative Results for small dataset.

	Number of vehicles	Time	Total distance	Cost
The proposed distribution scheme	6	2866	1072	3875.30
Compared distribution scheme	8	3137	1056	4528.72

TABLE 7. Comparative Results for large dataset.

	Number of vehicles	Time	Total distance	Cost
The proposed distribution scheme	23	35985	5922	49823.81
Compared distribution scheme	27	45724	5892	55246.19

TABLE 8. Comparative Results of the proposed scheme and algorithm using cplex for small dataset.

	Number of vehicles	Time	Total distance	Cost
The proposed distribution scheme	6	2866	1072	3875.30
Algorithm using CPLEX	6	2879	1075	3888.79

TABLE 9. Comparative Results of the proposed scheme and algorithm using cplex for large dataset.

	Number of vehicles	Time	Total distance	Cost
The proposed distribution scheme	23	35985	5922	49823.81
Algorithm using CPLEX	23	36016	5982	50001.62

V. CONCLUSION

The existing studies on the VRP rarely consider the actual road conditions. As a result, the routing plans for vehicles cannot achieve the desired outcome. To solve the problem, this paper puts forward a dynamic VRP model and solves it by the GA. It is assumed that the real-time traffic data are updated every 15mins in the transport network, and the customer demand is updated dynamically from time to time. The example analysis shows that my model and its route adjustment strategy can minimize the total transport cost by routing the vehicles from multiple depots under the time window constraints. The research findings help transport enterprises to make effective use of vehicles and receive more profits.

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SHAOQING ZHENG received the master's degree in management from Zhejiang Gongshang University, in 2018. He is an Associate Professor with the Hangzhou College of Commerce, Zhejiang Gongshang University. His research interests include national economic management, agricultural economy, logistics, and supply chain management.